FILMORA

SECURITY, PRIVACY & CONTENT PROTECTION SPECIFICATION

Project Name: FILMORA
Project Type: Free Online Movie Streaming Platform
Document Number: FILMORA-SEC-005
Document Version: 1.0
Status: Proposed Design
Backend: Node.js + Express.js
Database: PostgreSQL
Frontend: HTML5, CSS3 and JavaScript
Security Approach: Defense in Depth

TABLE OF CONTENTS

1. Introduction
2. Security Objectives
3. Security Architecture
4. User Account Security
5. Password Security
6. Authentication Security
7. Authorization and Access Control
8. Database Security
9. API Security
10. Frontend Security
11. Video Streaming Security
12. Content Protection
13. File Upload Security
14. Administrator Security
15. Privacy and Personal Data Protection
16. Session Management
17. Protection Against Common Attacks
18. Logging and Security Monitoring
19. Backup and Recovery Security

20. Incident Response
21. Security Testing
22. Deployment Security
23. Security Development Checklist

---

1. INTRODUCTION

Security is a fundamental requirement of FILMORA.

The platform will manage user accounts, personal information, movie information, streaming sessions and administrative operations.

This document defines the security and privacy requirements necessary to protect FILMORA and its users.

The security architecture must protect:

- User accounts.
- Passwords.
- Personal information.
- Database records.
- Movie files.
- Streaming sessions.
- Administrative operations.
- Platform infrastructure.

2. SECURITY OBJECTIVES

The primary objectives are:

1. Prevent unauthorized account access.
2. Protect sensitive user information.
3. Protect database access.
4. Secure communication between frontend and backend.
5. Prevent unauthorized movie access.
6. Protect administrator accounts.
7. Detect suspicious activities.

8. Maintain reliable backups.
9. Protect legally authorized movie content.
10. Support responsible personal data management.

3. SECURITY ARCHITECTURE

FILMORA will use a layered security architecture.

3.1 Security layers

Layer	Security responsibility
Frontend	Safe interface and input handling
API	Request validation and access control
Authentication	Verify user identity
Authorization	Check user permissions
Database	Protect stored information
Media storage	Restrict file access
Streaming	Authorize video delivery
Infrastructure	Protect servers and network

Each layer must operate independently while supporting the overall security architecture.

4. USER ACCOUNT SECURITY

FILMORA must protect registered accounts.

Requirements:

- Unique email addresses.
- Unique usernames.
- Email verification.
- Secure password storage.
- Account status management.
- Secure password recovery.
- Login attempt monitoring.
- Account deletion functionality.

Possible account statuses:

- Pending Verification.
- Active.
- Suspended.
- Disabled.
- Deleted.

5. PASSWORD SECURITY

Passwords must never be stored in plain text.

5.1 Password protection

The backend will use Argon2id for password hashing.

Requirements:

- Secure password hashing.
- Unique salts.
- Password strength validation.
- Protection against brute-force attempts.
- Secure password reset procedures.

5.2 Password requirements

FILMORA should require:

- A minimum password length.
- Protection against commonly used passwords.
- Clear password strength feedback.
- Secure password change procedures.

6. AUTHENTICATION SECURITY

The authentication system will manage:

- Registration.
- Login.
- Logout.
- Email verification.
- Password recovery.

- Session creation.
- Session expiration.
- Session revocation.

6.1 Session management

Sessions must:

- Use unpredictable identifiers.
- Be stored securely.
- Use HTTP-only cookies.
- Use Secure cookies over HTTPS.
- Apply appropriate SameSite settings.
- Expire after an appropriate period.
- Be revocable by the user or administrator.

7. AUTHORIZATION AND ACCESS CONTROL

FILMORA will implement Role-Based Access Control (RBAC).

7.1 System roles

Role	Access
Administrator	Full platform administration
Content Manager	Movie and media management
Moderator	Content moderation
Registered User	Personal account and streaming features

7.2 Authorization requirements

The backend must verify permissions before executing protected operations.

The frontend must never be treated as the authority for permissions.

Users must not be able to access administrative resources simply by changing a URL.

8. DATABASE SECURITY

PostgreSQL must be configured securely.

Requirements:

- Strong database credentials.

- Restricted database network access.
- Separate application database accounts.
- Least-privilege permissions.
- Encrypted connections.
- Secure backup storage.
- Parameterized queries.
- Regular database updates.
- Monitoring of database access.

Sensitive fields must be protected appropriately.

9. API SECURITY

All backend API requests must undergo validation.

Security requirements:

- HTTPS communication.
- Input validation.
- Rate limiting.
- Authentication checks.
- Authorization checks.
- CSRF protection where applicable.
- Secure CORS configuration.
- Consistent error handling.
- Request size restrictions.

Public API endpoints must expose only information intended for public access.

10. FRONTEND SECURITY

The frontend must:

- Avoid storing passwords.
- Avoid exposing API secrets.
- Avoid exposing database credentials.
- Avoid inserting untrusted HTML.
- Validate form inputs.

- Handle expired sessions.
- Use HTTPS.
- Display safe error messages.

Frontend validation must always be repeated by the backend.

11. VIDEO STREAMING SECURITY

The streaming system must protect access to authorized movie files.

Requirements:

- Authentication before protected playback.
- Verification of movie availability.
- Verification of content permissions.
- Short-lived streaming authorization.
- Restricted media storage.
- Secure CDN configuration.
- Expiration of access credentials.
- Monitoring of suspicious access.

The backend must not expose private storage credentials or permanent unrestricted media links.

12. CONTENT PROTECTION

FILMORA will only publish movies for which the platform has appropriate streaming and distribution rights.

12.1 Content authorization

Before publishing a movie, the administration system must verify:

- Content ownership or licensing.
- Authorized territories.
- Licensing start date.
- Licensing expiration date.
- Authorized distribution method.

12.2 Content protection mechanisms

Depending on the licensing requirements, FILMORA may implement:

- Signed media URLs or cookies.

- CDN access restrictions.
- Digital Rights Management (DRM).
- Playback session monitoring.
- Visible or forensic watermarking where appropriate.

No technical protection method guarantees that streamed content cannot be copied.

13. FILE UPLOAD SECURITY

All uploaded files must be validated.

Requirements:

- File type validation.
- File size restrictions.
- Secure file naming.
- File scanning where appropriate.
- Restricted storage permissions.
- Upload authorization.
- Processing isolation.
- Upload activity logging.

Uploaded files must not be executed as application code.

14. ADMINISTRATOR SECURITY

Administrator accounts require additional protection.

Requirements:

- Strong passwords.
- Multi-factor authentication.
- Restricted administrative access.
- Role-based permissions.
- Administrative activity logs.
- Secure session management.
- Account suspension controls.

Administrative operations must be auditable.

15. PRIVACY AND PERSONAL DATA PROTECTION

FILMORA will collect only the information required to operate the platform.

Potential personal information includes:

- Email address.
- Username.
- Profile information.
- Account activity.
- Viewing history.
- User preferences.
- Security session information.

Privacy requirements:

- Provide a privacy policy.
- Explain what information is collected.
- Explain why information is collected.
- Restrict access to personal information.
- Provide account deletion procedures.
- Define data retention periods.
- Protect personal information during transmission and storage.

Applicable privacy legislation must be identified according to the territories where FILMORA operates.

16. SESSION MANAGEMENT

Users must be able to manage their active sessions.

Features:

- View active sessions where supported.
- Log out.
- Revoke individual sessions.
- Revoke all sessions.
- Expire inactive sessions.
- Invalidate sessions after sensitive account changes.

17. PROTECTION AGAINST COMMON ATTACKS

FILMORA must be designed to mitigate the following threats:

Threat	Protection
SQL Injection	Parameterized queries
Cross-Site Scripting	Output encoding and safe rendering
Cross-Site Request Forgery	CSRF protection
Brute-force login	Rate limiting
Account takeover	Secure authentication and MFA
Unauthorized access	Access control
Session theft	Secure cookies and HTTPS
Malicious uploads	File validation and scanning
Credential exposure	Secret management
Unauthorized media access	Restricted storage and authorization

18. LOGGING AND SECURITY MONITORING

The platform should record:

- Failed login attempts.
- Successful authentication events.
- Password changes.
- Account security changes.
- Administrative actions.
- Suspicious API activity.
- Streaming authorization failures.
- File upload failures.
- Database errors.

Logs must not contain passwords, session tokens or private credentials.

19. BACKUP AND RECOVERY SECURITY

Backup requirements:

- Encrypted backups.

- Restricted backup access.
- Automated backup schedules.
- Defined retention policies.
- Regular recovery testing.
- Secure backup storage.
- Separate backup credentials.

The database and media storage must have independent backup strategies.

20. INCIDENT RESPONSE

FILMORA must establish a security incident response procedure.

20.1 Incident stages

1. Detection.
2. Investigation.
3. Containment.
4. Eradication.
5. Recovery.
6. Post-incident review.

Examples of incidents:

- Unauthorized account access.
- Database exposure.
- Suspicious administrator activity.
- Media storage compromise.
- Malware upload.
- Unauthorized data disclosure.

Affected users and relevant authorities must be notified where required by applicable law.

21. SECURITY TESTING

The platform must undergo:

- Authentication security testing.
- Authorization testing.
- API security testing.

- Database security testing.
- File upload testing.
- Session management testing.
- Streaming access testing.
- Dependency vulnerability scanning.
- Penetration testing.
- Backup restoration testing.

22. DEPLOYMENT SECURITY

Production deployment must include:

- HTTPS certificates.
- Secure environment variables.
- Restricted server access.
- Updated dependencies.
- Firewall configuration.
- Database access restrictions.
- Secure storage configuration.
- Application monitoring.
- Automated backups.

Development credentials must never be reused in production.

23. SECURITY DEVELOPMENT CHECKLIST

Before FILMORA goes live, the following must be completed:

- Password hashing implemented.
- Registration secured.
- Login secured.
- Email verification implemented.
- Secure sessions implemented.
- Role-based authorization implemented.
- Database access restricted.
- API validation implemented.

- Rate limiting configured.
- HTTPS enabled.
- Secure video delivery configured.
- Content rights verified.
- File upload security implemented.
- Administrator MFA implemented.
- Privacy policy published.
- Backup system tested.
- Security testing completed.

FINAL SECURITY PRINCIPLE

FILMORA will use a defense-in-depth security architecture to protect user accounts, personal information, administrative resources, database records and authorized movie content.

Security must be incorporated throughout development rather than added only after the platform has been completed.

DOCUMENT STATUS: PROPOSED — AWAITING APPROVAL

FILMORA — YOUR WORLD OF MOVIES.